Mustafa Mehmood

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Experience

Applied Scientist — *London, UK*
Thomson Reuters (CoCounsel, AI legal assistant) — *June 2025 – Present*

- Developed core agentic workflows for M&A due-diligence review across hundreds of documents per deal, grounding agents via tools in expert-curated legal sources; achieves 90%+ recall of lawyer-flagged material risks.
- Own evaluation methodology: defined grading criteria with lawyers and designed LLM-as-judge strategies calibrated to expert scores (0.78 Cohen's κ), detecting 87.6% of human-flagged errors; cut eval cycles from days to hours.
- Built an orchestrator agent harness and skill library encoding lawyers' review methodology, plus MCP tools for citation verification, cross-reference resolution and redline comparison.
- Compared context-engineering strategies for long documents (retrieval-then-reason, progressive summarization, sub-agent state passing), yielding up to 4× fewer orchestrator compactions without loss in recall.
- Fine-tuned Qwen-32B for clause customization with QLoRA (SFT, then DPO on lawyer preference rankings), matching GPT-4o in lawyer review at 5×+ lower per-token cost.

LLM Data Integrity Specialist — *Nottingham, UK*
Outlier AI — *May 2024 – May 2025*

- Improved tool use in frontier models through RLHF training pipelines, enabling them to chain and execute multi-step actions from high-level user intent, supporting AI development at firms including Google and Meta.
- Curated instruction tuning and preference datasets to drive a 20% improvement in instruction adherence.

Data Scientist Intern (Part-Time) — *Selangor, Malaysia*
Department of Neuroscience, University of Nottingham Malaysia — *June 2021 - Aug 2023*

- Engineered feature extraction pipelines to transform EEG and eye-tracking clinical data into structured features: coherence, Hjorth parameters, and spectral power bands for EEG; gaze patterns and fixation analysis for eye-tracking.
- Built diagnostic classifiers reaching 80% accuracy for Parkinson's and 82% for Schizophrenia from EEG signals, and 92% for autism by representing gaze scanpaths as images and training a custom transfer-learning computer vision model.

Education

University of Nottingham — *Nottingham, UK*
MSc *Artificial Intelligence* - ***Grade: Distinction*** — *Sept 2023 – Dec 2024*
BSc *Computer Science* ***(Hons)*** — *Sept 2020 – Aug 2023*
Key Modules: Machine Learning, Computer Vision, Natural Language Processing, Big Data, Information Visualisation

Key Projects

Sherlock-AI: Character Emulation with LLMs — Website Repo

- Designed a resource-efficient, transferable character emulation pipeline: fine-tuned a quantized LLaMA model with *LoRA* for Sherlock Holmes emulation, automating dialogue extraction and synthetic data generation from public domain texts.
- Implemented ReAct-style reasoning for multi-step deductive analysis with dual inference support (OpenRouter API and local GGUF), optimizing for performance and cost across cloud and local setups.
- Built an interactive story mode where users play as characters in branching Sherlock Holmes mysteries through structured decisions and free-form actions, using templated prompts with streaming structured output to deliver real-time scene generation and auto-generated context summaries.
- In A/B testing, 72.4% of users favoured the fine-tuned model over baseline for authenticity, with perfect usability scores.

CourseCompanion: RAG-Powered Course Assistant — Repo

- Developed an educational platform enabling professors to create AI teaching assistants that auto-generate knowledge bases from course materials, featuring chat interfaces, FAQ system, and management dashboards; built with Flask and PostgreSQL using FAISS-backed RAG with LangChain and Flair.

DeepGaze: Autism Diagnosis with ML and Webcam Eye-Tracking — Demo Video Repo

- Built a modular Flask web app for webcam-based eye-tracking data collection, visualization, real-time prediction, and model retraining, operationalizing the autism-detection method from my publication.

Publication

Mehmood, M., Amin, H. U. (2024). Pre-diagnosis for Autism Spectrum Disorder Using Eye-Tracking and Machine Learning Techniques. *Advances in Brain Inspired Cognitive Systems*, Springer Nature Singapore.

Technical Skills

Programming Languages: Python, C, Java, JavaScript, SQL, HTML/CSS
Libraries & Frameworks: PyTorch, TensorFlow, Hugging Face (Transformers, trl), scikit-learn, LangChain, LangGraph, Pydantic AI, spaCy, vLLM, FastAPI, Flask, pandas, NumPy, FAISS, Weights & Biases, Gradio/Streamlit, Claude Agent SDK
Tools & Platforms: Git/GitLab CI, Docker, AWS (SageMaker, EC2, Lambda, Bedrock), Spark/Databricks, PostgreSQL, Pinecone, Runpod, DigitalOcean, MCP, Agent Harness Design, Claude Code, Cursor